

Brief Contents

Chapters

1. Introduction to Computer Architecture & Organization
2. History of Microprocessors & Microcontrollers
3. PIC 18 Microcontroller Architecture & Features
4. Instruction Set, Addressing Modes, Directives & Tools
5. Programming PIC 18 using Assembly & C Language
6. Overview of Peripherals & Interfacing
7. PIC 18 I/O Ports & Interrupts
8. Flash, EEPROM & Interfacing of memories
9. PIC 18 Timers & its Programming
10. PIC 18 Parallel Slave Port, Serial port & its programming
11. Interfacing Displays & Keyboards
12. Interfacing Sensors, ADC, DAC & Comparators
13. Programming CCP & ECCP & PWM
14. Interfacing Actuator's – Relays, DC, Servo & Stepper Motors

Contents

Chapters

1. Introduction to Computer Architecture & Organization
 - 1.1 Digital Electronics Fundamentals
 - 1.2 Functional units of a computer.
 - 1.3 Bus structure
 - 1.4 Memory Organization
 - 1.5 Von Neumann & Harvard Architectures
 - 1.6 Basic Addressing modes, Instruction formats & Size
 - 1.7 Operational Concept of a computer
 - 1.8 Register Bus Structure
 - 1.9 Configuration and design of ALSU
 - 1.9.1 Configuration and design of ALU
 - 1.9.2 Configuration and design of Shift/Rotate Unit
 - 1.9.3 Configuration and design of combined ALU, Shift/Rotate and multi bus structure unit.
 - 1.10 Optimization
 - 1.11 Timing & Control Unit
 - 1.12 Pipelining
2. History of Microprocessors & Microcontrollers
 - 2.1 History of Microprocessors
 - 2.1.1 Intel Processors
 - 2.1.2 ARM Processors
 - 2.2 Introduction to 8085
 - 2.3 Introduction to 8086
 - 2.4 Microprocessors vs Microcontrollers
 - 2.5 Introduction to 8051
 - 2.6 History of Microchips Microcontrollers
 - 2.6.1 PIC16 Series
 - 2.6.2 PIC18 Series

- 2.6.3 PIC24 Series
 - 2.6.4 PIC32 Series
 - 2.6.5 SAM Series
 - 2.6.6 Other Microchip Series – Atmel, AVR & Other
- 3. PIC 18 Microcontroller Architecture & Features
 - 3.1 PIC 18 Family Overview
 - 3.2 Comparison of 8051 & PIC18F458
 - 3.3 Features of PIC 18F458
 - 3.4 Architecture of PIC 18F458
 - 3.5 Memory organization
 - 3.5.1 ROM Organization
 - 3.5.2 Stack Memory
 - 3.5.3 RAM Organization
 - 3.6 Instruction Cycle and pipelining
 - 3.7 Clock Configuration
- 4. Instruction Set, Addressing Modes, Directives & Tools
 - 4.1 Data Formats
 - 4.2 Labelling Rules
 - 4.3 Assembler Directives
 - 4.4 Addressing Modes
 - 4.5 PIC 18 Instruction Set
 - 4.5.1 Data Transfer
 - 4.5.2 Arithmetic
 - 4.5.3 Logical
 - 4.5.4 Bit-Wise
 - 4.5.5 Branch Transfer
 - 4.5.6 Table/ROM Program Memory
 - 4.6 Overview of MPLAB Simulator
- 5. Programming PIC 18 using Assembly & C Language
 - 5.1 Assembly Language Programming Examples
 - 5.1.1 Basic programming examples
 - 5.1.2 Bank Switching

- 5.1.3 Macros & Modules
- 5.2 C Language Programming Examples
- 6. Overview of Peripherals & Interfacing
 - 6.1 Introduction to communication
 - 6.2 Introduction to Parallel communication
 - 6.3 Introduction to Serial Communication
 - 6.4 RS 232 Standard
 - 6.5 Voltage level converters for RS 232 to TTL, CMOS
 - 6.6 Interfacing Signals
 - 6.7 Memory Interfacing concepts
 - 6.7.1 Memory Segmentation
 - 6.7.2 Memory/Register Banks
 - 6.7.3 Memory Interleaving
 - 6.7.4 Mapping Techniques
 - 6.7.4.1 Memory mapped I/O
 - 6.7.4.2 I/O mapped I/O
 - 6.7.5 Memory Interfacing
 - 6.8 Traditional Peripheral devices
 - 6.8.1 Programmable Peripheral Interface – 8255
 - 6.8.2 USART – 8251
 - 6.8.3 Programmable Interrupt Controller – 8259
 - 6.8.4 Direct Memory Access – 8237
- 7. PIC 18 I/O Ports & Interrupts
 - 7.1 IC packages & Pin diagram of PIC18 Devices
 - 7.2 I/O Port Structure
 - 7.2.1 Overview of Functional Modules & its Pins
 - 7.2.2 Port A
 - 7.2.3 Port B
 - 7.2.4 Port C
 - 7.2.5 Port D
 - 7.2.6 Port E
 - 7.3 Interfacing & Programming of LED's

- 7.4 Interfacing & Programming of Switches
- 7.5 Overview of Interrupts & its Structure
 - 7.5.1 Registers related to Interrupts
- 7.6 External Interrupts & its programming
 - 7.6.1 Interrupt on Change/PORTB Change
 - 7.6.2 External Interrupts
- 8. Flash, EEPROM & Interfacing of memories
 - 8.1 Table Pointer & Operations
 - 8.2 Flash Memory Operations – Reading, Writing/Erasing
 - 8.3 EEPROM Memory Operations – Reading, Writing/Erasing
 - 8.4 External Memory Interface
 - 8.5 Interfacing of External Memories - PIC18C801/PIC18F8720
- 9. Timers & its Programming
 - 9.1 Overview of Timers
 - 9.2 Timer structures & its programming
 - 9.2.1 Timer 0
 - 9.2.2 Timer 1
 - 9.2.3 Timer 2
 - 9.2.4 Timer 3
 - 9.2.5 Timer 4
 - 9.3 Using Timers as counters
 - 9.4 Using Timers to determine number of instruction cycles
- 10. PIC18 Parallel Slave Port, Serial Port & its Programming
 - 10.1 Parallel Slave Port
 - 10.2 Introduction to PIC18 Serial Communication
 - 10.3 USART Asynchronous Mode
 - 10.3.1 Transmit
 - 10.3.2 Receive
 - 10.4 USART Synchronous Mode
 - 10.4.1 Transmit
 - 10.4.2 Receive

- 11. Interfacing Displays & Keyboards
 - 11.1 Introduction to Displays
 - 11.2 Interfacing Seven Segment Display
 - 11.3 Interfacing LCD Display
 - 11.4 Interfacing Keypad
- 12. Interfacing Sensors using ADC, DAC & Comparators
 - 12.1 ADC & Interfacing with Sensors
 - 12.2 PIC18's 10-Bit ADC & Configuration
 - 12.3 Comparators & Configuration
 - 12.4 DAC & Interfacing
 - 12.5 PIC18's 5-Bit DAC & Configuration
- 13. Programming CCP & ECCP & PWM
 - 13.1 Overview of CCP
 - 13.2 CCP Modes of Operation
 - 13.2.1 Compare
 - 13.2.2 Capture
 - 13.2.3 PWM
 - 13.3 Using ECCP
- 14. Interfacing DC Motor, Stepper & Servo Motors
 - 14.1 Interfacing of DC Motor
 - 14.2 Interfacing of Stepper Motor
 - 14.3 Interfacing of Servo Motor

